Examiner's Use

2 **(a)** State what is meant by *electric field strength*.

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.....

.....

..... [2]

(b) In a crystal of sodium chloride, a sodium ion of charge $+e$ is separated from its closest chloride ion of charge $-e$ by a distance of 2.8×10^{-10} m.

Assume that the ions are point charges.

(i) Calculate the magnitude of the electric field strength that one sodium ion creates at its closest chloride ion.

electric field strength = ... N C^{-1} [2]

(ii) Calculate the attractive force between the two ions.

force = ... N [1]

(c) The force in **2(b)(ii)** does not accelerate the sodium and chloride ions towards each other due to presence of other ions.

Figure 2.1 represents a plane of sodium and chloride ions at their equilibrium distance.

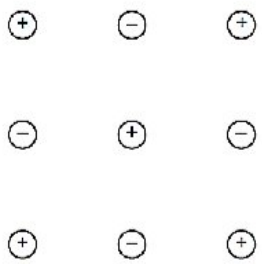


Figure 2.1

(i) With reference to Figure 2.1, state and explain the resultant force on the central ion.

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..... [3]

(ii) The central ion vibrates out of position. State the effect of the resultant force on the ion.

..... [1]

(iii) Suggest what affects the equilibrium distance between any two adjacent sodium and chloride ions.

.....

..... [1]

Examiner's Use

(d) Sodium chloride and potassium chloride are both essential mineral salts that act as vital electrolytes in the human body. They have similar physical properties and potassium chloride is often used as a sodium-free salt substitute to help lower blood pressure.

The equilibrium distance for sodium chloride and potassium chloride is 0.28 nm and 0.31 nm respectively.

Suggest, with a reason, which of the two salts has a higher melting point.

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..... [2]

[Total: 12]